

**Understanding the Access to Food and Medical Resources Inside
Washington Latino Communities**

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1- Introduction/Background

The Latino Educational Training Institute (LETI) is dedicated to fostering financial and personal success for Latino immigrants and low-income individuals. They support families across all areas including basic needs, health and wellness, education, and cultural awareness. LETI plans to expand its services to Everett Washington. So as participants in the Data for Public Good program, we aim to utilize LETI's data and public resources to offer insights and recommendations to assist them in their move. We hope to bring in new perspectives and help them develop even more comprehensive programs to serve the community. Our main objective is to focus on understanding the access to food and medical resources inside Latino communities in the greater Seattle area.

Goals:

- Create geographic maps with the locations of the families and individuals struggling with food or healthcare insecurity or both and suggest solutions to improve their conditions.
- Identify relationships between people under the poverty line and people struggling with both food insecurity and healthcare accessibility.
- Discover potential alignment between community resource locations and people who are struggling with accessing community resources.

2 - Datasets

The datasets our group focused on are the ones collected by LETI during 2021 and 2023. These datasets were given to us by the non-profit organization LETI (Latino Educational Training Institute). The data was also collected and gathered by LETI and was funded by Amerigroup Health Care, the Department of Financial Institutions, the WA State Department of Commerce, and many more. The specific datasets we focused on are listed below:

Datos Demograficos de Clientes (Customer Demographic Data)

This dataset helped our team with finding the specific demographics of LETI families and individuals who are specifically struggling with food and healthcare availability. We utilized this dataset's variables of zip code, and poverty level as well during our analysis. The variables used from the dataset are as follows: "Are you worried that you or someone else in your household doesn't have enough food to eat?", "Do you need access to a primary care physician?", "Do you need help with behavior, mental health, or substance use?", and "Do you need access to dental care?". We later decided to bundle all the health-related variables into one for further analysis. This variable was named "Need Access/Assistance in at least one type of medical resources". This dataset has 37 variables and 195 cases.

Unduplicated Clients @ CDO Event (Community Event Survey)

This dataset helped us identify the location and the number of people who need assistance in finding community and medical resources and healthcare services in Snohomish County. It has 13 variables and a total of 409 cases, the variables we used are zip code, "What is your first language?", "Country of origin?", "How long have you lived in Washington State?", and "Do you have a hard time accessing or finding community and medical resources in person or online because of language, distance, or cultural barriers?".

Limitations

A few limitations we noticed about the LETI datasets from 2021 and 2023. Both datasets have limited questions on the demographics and financial status. These variables would have provided our team with a more comprehensive understanding of the respondents' needs and challenges. Additionally, there is a lack of parity between the 2021 and 2023 datasets due to the differing in focus of different programs. This led to inconsistencies in the variables and data collected, preventing us from conducting an in-depth analysis between the two years. As a result, tracking program progress and identifying possible trends become challenging.

3 - Data Story

2021 LETI Survey Data

The 2021 LETI survey data stood out to us for having an ample amount of datasets to work with. With this wide variety of data, our team had to narrow the scope to specific health and food-related datasets. That is why our team felt it was important to use the dataset "Datos Demograficos de Clientes" which focused on surface-level demographic information with a few additional questions pertaining to the conditions of the respondents.

This first visualization was designed to help us understand the locations of the individuals struggling with finding access to food at the city level. Though the dataset provided additional

cities there weren't enough respondents to make it statistically relevant (minimum 20 responses). This figure was also replicated exactly so it could match the next visualization on the individuals struggling with getting access to medical resources. Both figures seem to show a high percentage of struggle across every location, having all cities struggling with food or medical at very high percentages. This shows us that a lot of work needs to be done across the board to help assist these Latino Families and individuals. The City that seems to be struggling the most is Lynnwood, with a collective food insecurity score of around 84% and a medical insecurity score of around 80%.

Figure 1. A graph depicting the 'struggling with food insecurity' variable as a stacked bar chart showing only the cities most recorded from the LETI 2021 dataset (min. 20 responses). The red indicates struggling while the light grey indicates not struggling.

Struggling with Food Insecurity

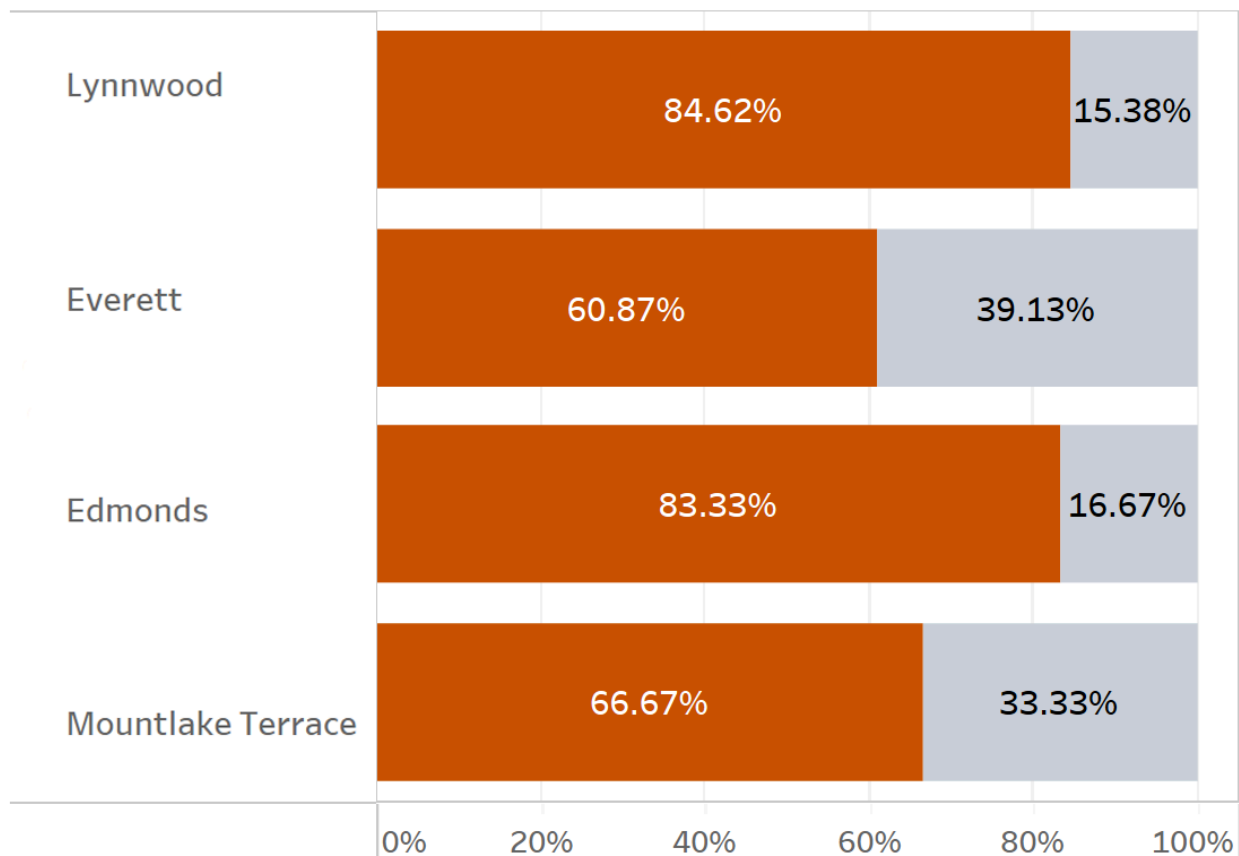
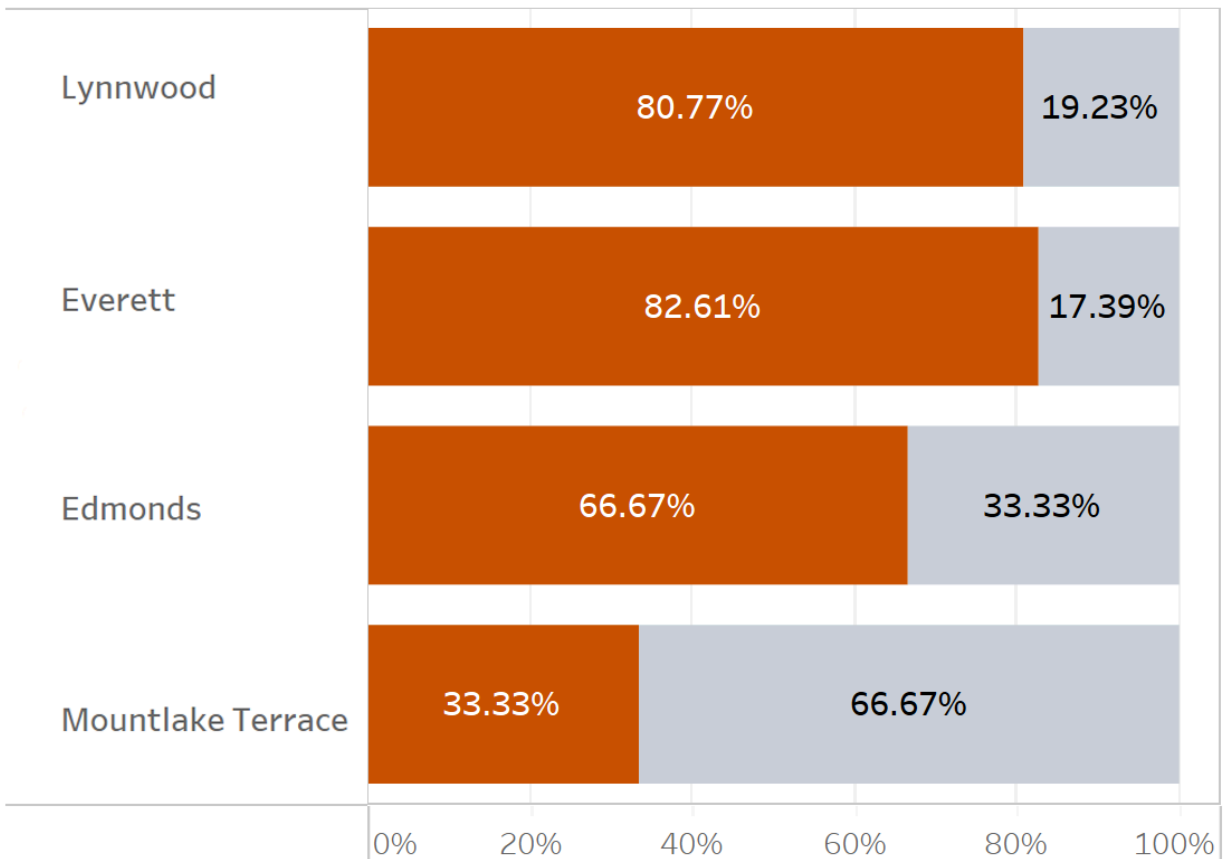


Figure 2. A graph depicting a stacked bar chart of the cities most recorded from the LETI 2021 dataset (min. 20 responses) that reported struggling with getting any sort of medical assistance. The red indicates struggling while the light grey indicates not struggling.

Struggling with getting Medical Assistance



The next graph was made to help visualize the respondents by zip code, showing which areas struggle with both medical and food insecurity at the same time. The graph is laid out as a map depicting both unique colours and symbols. The colored regions represent the majority of respondents in that region who reported either struggling with getting access to food or not struggling. The symbols represent the majority of respondents in that region who reported either struggling with getting access to medical resources or not struggling, just like the food variable spoken about previously. This graph shows us a common trend that most regions that respond with a high percentage of struggle for either one of the variables seem to be struggling with the other. There only seem to be 2 out of the 19 zip codes where this does not apply.

Figure 3. A graph depicting a map graph that visually represents respondents' struggle with both medical and food insecurity by zip code. The graph uses unique colours and symbols to differentiate areas where respondents reported difficulty in accessing food or medical resources, indicating a strong correlation between the two types of insecurities.

Do You Struggle with Food Availability and Medical Assistance? Divided by Zip Code.

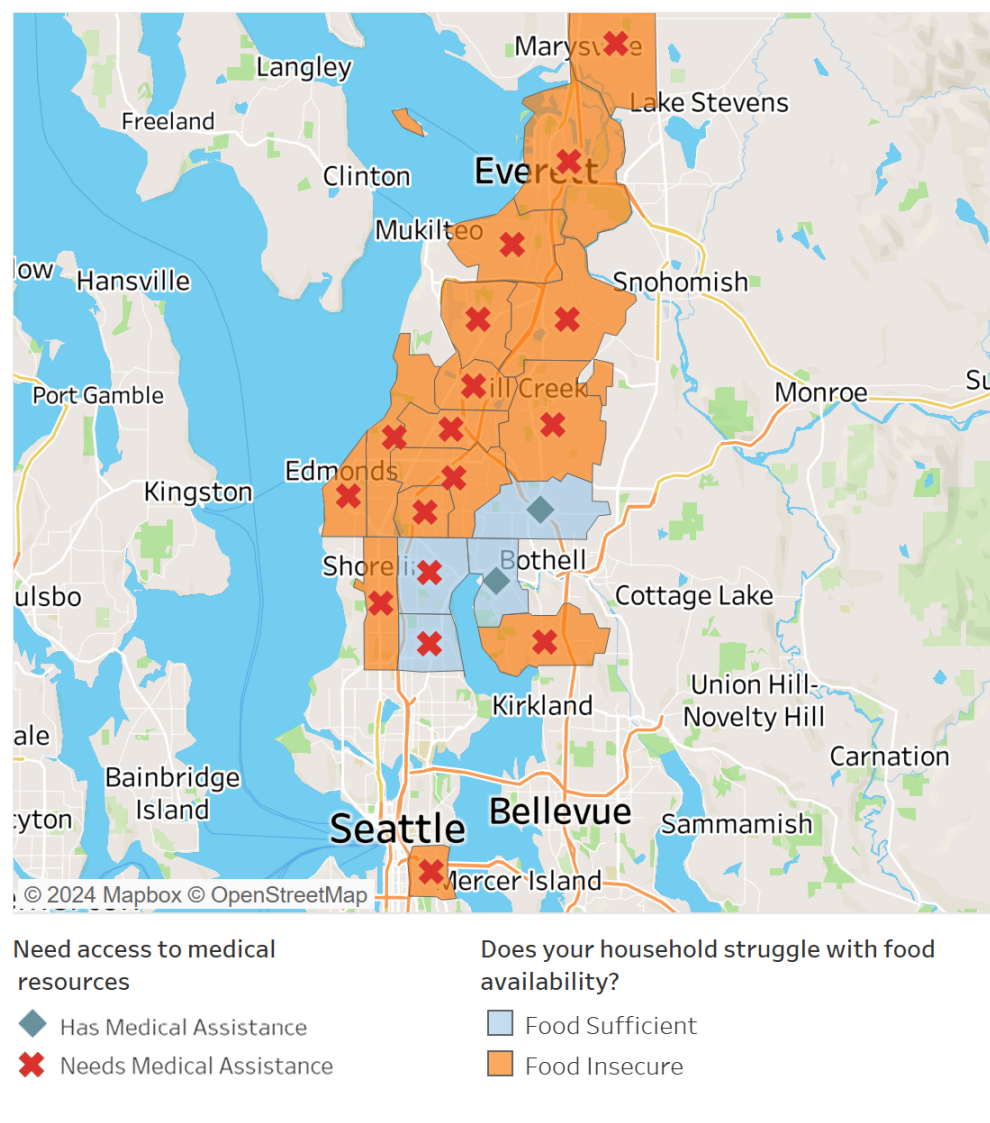
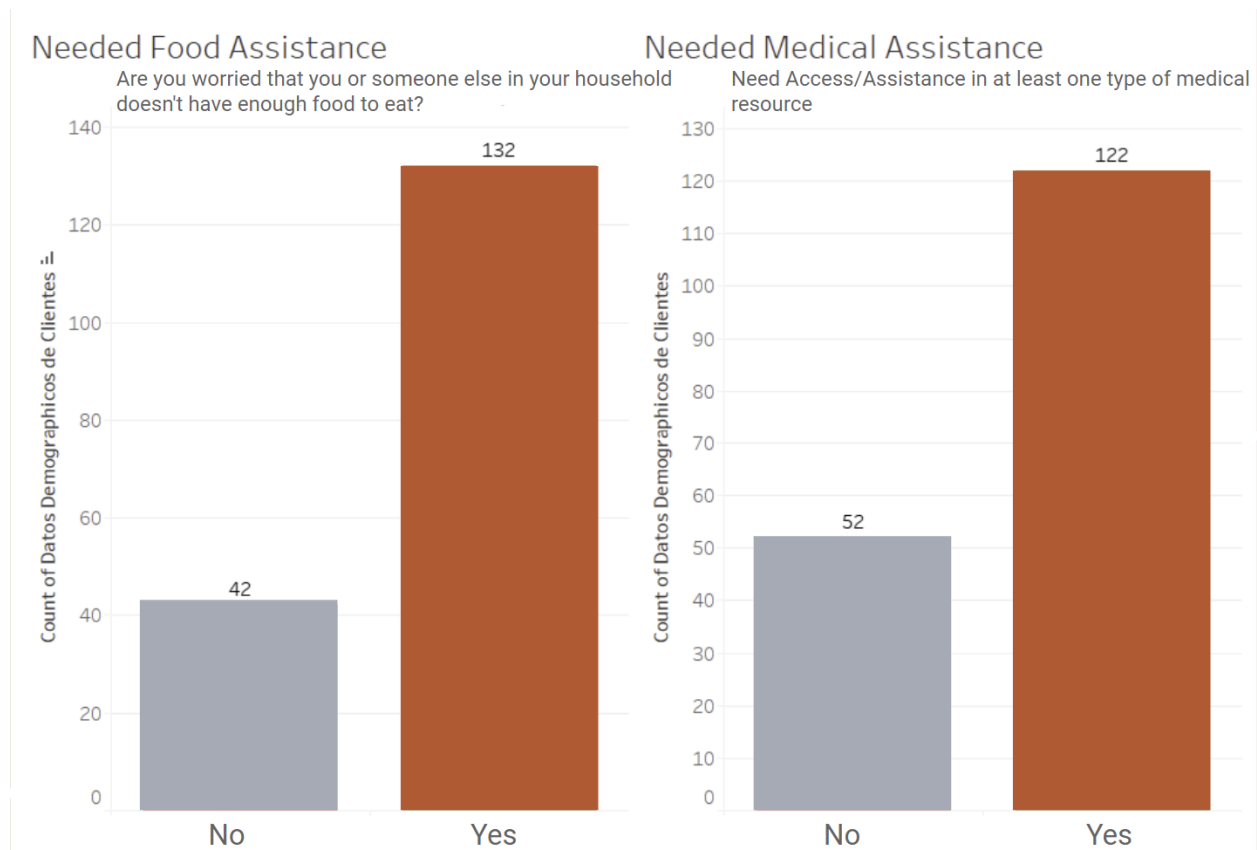


Figure 4. A graph depicting the need for medical assistance and food assistance. The analysis aims to depict the correlation between the need for these two types of assistance among respondents from various regions.



This graph shows the proportion of respondents requiring medical and food assistance, emphasizing the areas with the highest reported needs. The visualization indicates a strong correlation between the need for medical and food assistance, underscoring the compounded challenges faced by certain communities in accessing essential resources. To be more specific the needed food assistance and needed medical assistance variables show a strong correlation of about 94%. This data can inform targeted support measures to address the dual needs of these populations.

2023 LETI Survey Data

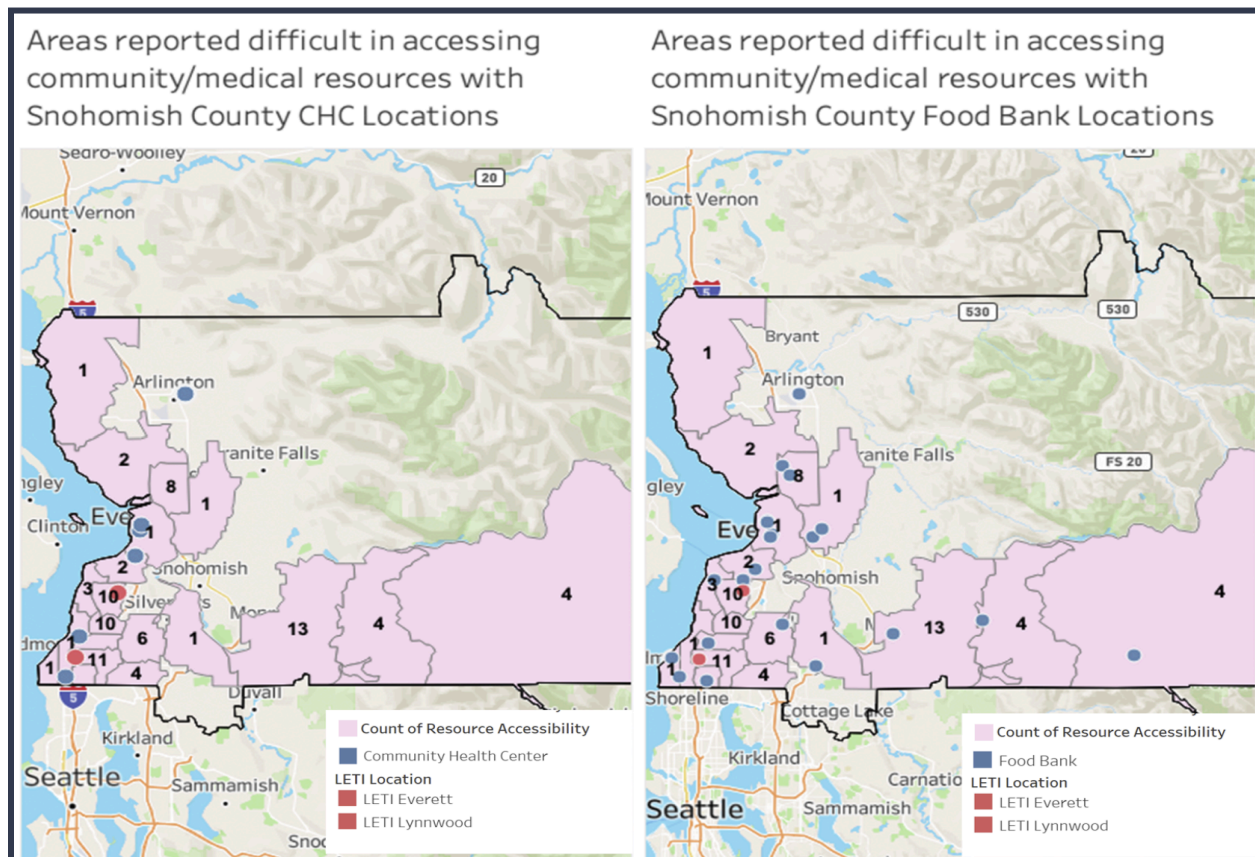
From the LETI 2023 survey dataset, we focused on one of the survey questions, “Do you have a hard time accessing or finding community and medical resources in person or online because of language, distance, or cultural barriers?”. The respondents were given a 1 to 5 scale option indicating the difficulty level, where 1 indicates not difficult at all and 5 indicates very difficult. We converted the 5-point scale into three categories: easy (1 and 2), medium (3), and difficult (4 and 5).

The two maps in Figure 1. focus on zip code areas where respondents reported difficulty in finding community and medical resources, a rating of either 4 or 5 on the survey, along with the corresponding respondent counts for each zip code. The LETI locations are represented as red dots on the maps, with blue dots indicating Community Health Center (CHC) and food bank

locations for comparison.

Based on Figure 1. Food bank locations are widely spread across nearly all areas where respondents reported difficulty accessing resources. In contrast, CHC locations are condensed and mainly situated in Everett, Lynnwood, and Edmonds. As a result, transportation and distance may become the main issues for people trying to access medical resources.

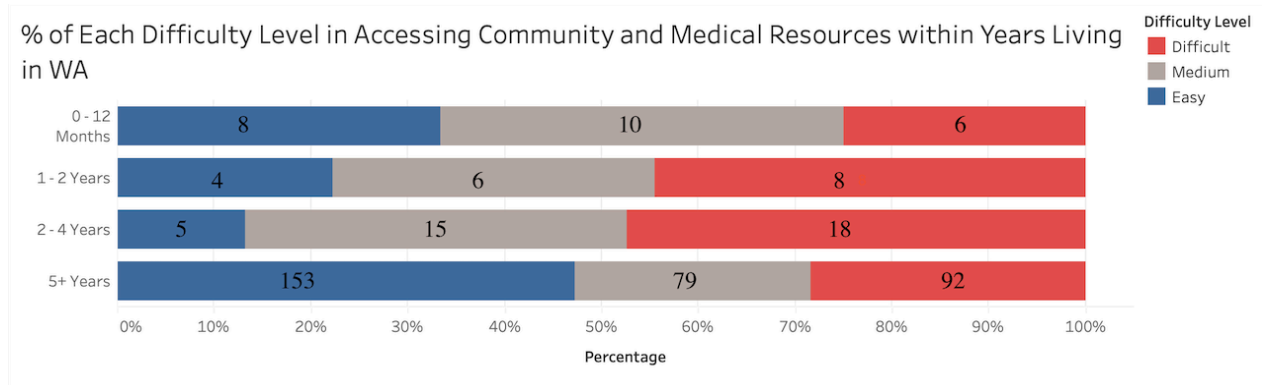
Figure 5. Areas reported difficulty in accessing community and medical resources within Snohomish County with CHC and food bank locations.



One additional variable we used to create the stacked bar chart in Figure 2. was “How long have you lived in Washington State?”. Respondents were given five options, 0 to 12 months, 1 to 2 years, 2 to 4 years, and 5 or more years. Each bar represents one of these groups while the color in each bar indicates the percentage of each difficulty level in accessing community and medical resources within each group.

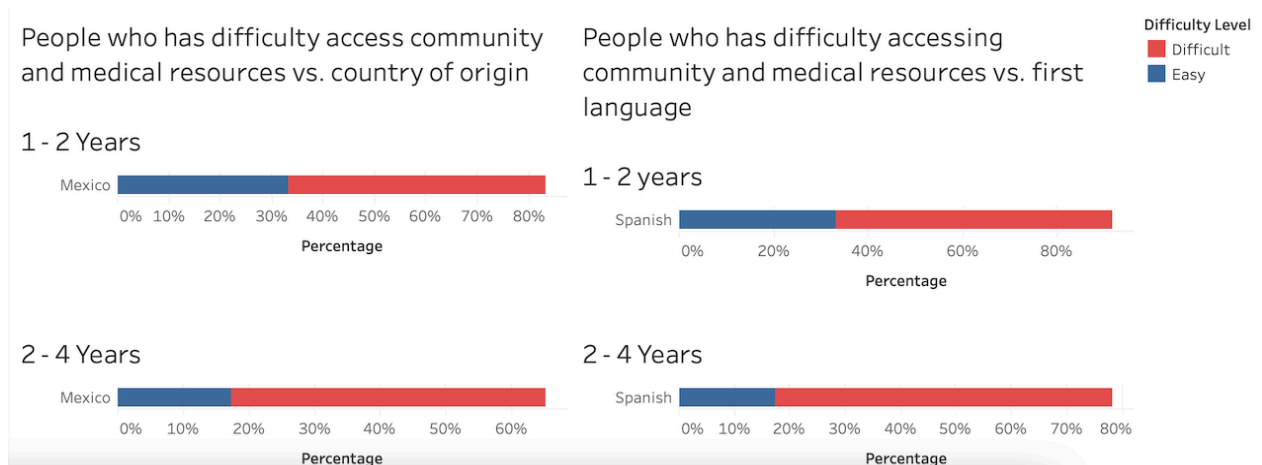
The percentage of people reporting difficulty accessing these resources in the 1 to 2 years and 2 to 4 years groups is significantly higher than the other two groups, with about 40% to 50% of respondents within these groups experiencing difficulty. However, a significantly higher percentage of respondents in the 5 or more years group found it easy to access resources, suggesting that the situation stabilizes over time.

Figure 6. Percentage of each difficulty level in accessing community and medical resources within years living in Washington state.



We extracted the easy and difficult data from 1 to 2 years and 2 to 4 years groups from Figure 2. to further investigate other demographic variables including “Country of origin?” and “What is your first language?” in Figure 3. We noticed that since LETI primarily targets the Latino community, the majority of the respondents are Spanish speakers and immigrants from Mexico. Compared to those who reported easy access to resources, a higher proportion of respondents sharing these demographic characteristics reported difficulty.

Figure 7. Respondents who either reported easy or difficulty in accessing community and medical resources that are living in Washington state for 1 - 2 years and 2 - 4 years based on country of origin and first language.



4 - Summary/Conclusion

The analysis of LETI’s 2021 and 2023 survey data highlights a struggle to access food and medical resources within Latino communities in Washington state. High percentages of respondents across various cities reported facing these challenges, with Lynnwood being notably impacted. The data also reveals a strong correlation between food insecurity and medical

insecurity, indicating that these struggles seem to be faced by similar individuals and families. Our data also revealed that newer residents to the US and those who are Spanish-born speakers or immigrants from Mexico, experience greater difficulty in accessing essential resources. This suggests a need for more programs to help assist the immigrant communities who have been living in the US for around 2-4 years.

Furthermore, the concentration of Community Health Centers (CHCs) in areas like Everett, Lynnwood, and Edmonds suggests that the locations of these centers are not the issue but transportation or lack of awareness as the main barriers. To address these issues, LETI could consider strategies such as mobile health clinics, partnerships with local transportation services, and increased outreach in highly affected areas. Additionally, providing more comprehensive support to newer immigrant residents can help stabilize their access to resources over time. By focusing on these measures, LETI can better serve the Latino community, and foster greater health to the Latinos of Washington.

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